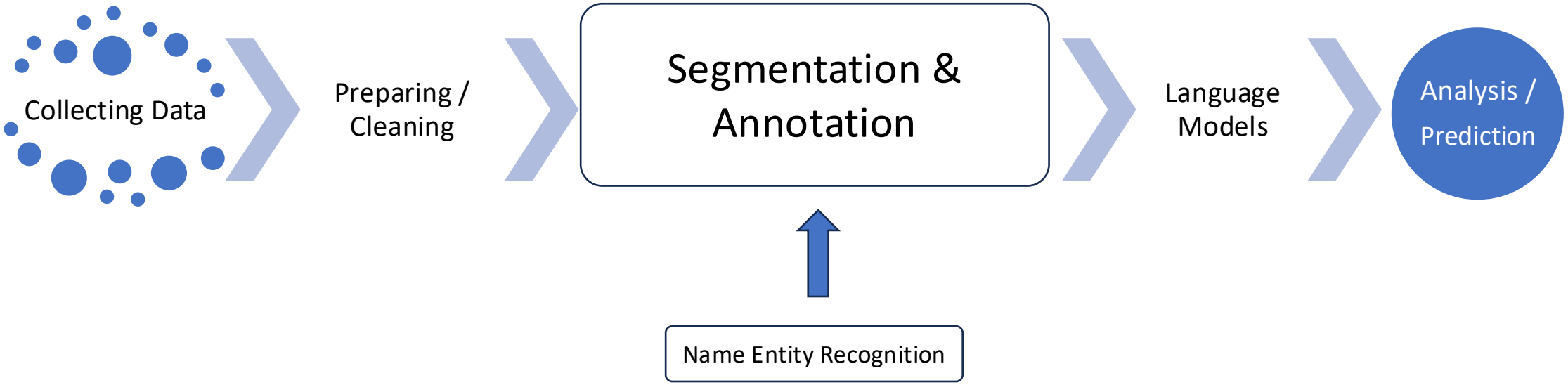
Week 19: Named Entity Recognition

SCC453 Natural Language Processing and Language Models

Ignatius Ezeani

NLP Overview / Pipeline



Learning Outcomes

By the end of the session, you should be able to:

- define named entity recognition and distinguish entity detection from entity linking
- explain IO/BIO/BILOU tagging schemes and why span boundaries matter
- compare rule-based, feature-based, neural, and transformer and LLM-based NER systems
- describe common sources of error:
  - *ambiguity*, *domain shift*, *nested entities*, and *unseen names*
- evaluate NER using entity-level precision, recall, and F1
- design a small NER experiment and interpret model errors responsibly

So, What is Named Entity Recognition?

- A **named entity** is "*anything that can be referred to with a proper name: a person, a location, an organization*" - Jurafsky & Martin (2026)
- Can be a Multiword Expression (MWE) e.g. the name "**Marie Curie**", the location "**New York City**", or the organization "**Lancaster University**"
- **Named Entity Recognition** (NER) is a ***sequence labelling*** task to assign words or phrases with tags like:
  - PERSON (PER), LOCATION (LOC), ORGANIZATION (ORG), or GEO-POLITICAL ENTITY (GPE)
- It is often a first step towards other NLP tasks:
  - sentiment analysis (e.g. identifying product names)
  - linking text to structured knowledge source e.g. Wikipedia
- In practice it is extended to other similar areas

Jurafsky & Martin, 2026, Speech and Language Processing (Draft) January 6, 2026. Chapter 17 Sequence Labeling for Parts of Speech and Named Entities

So, What is Named Entity Recognition?

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NER finds spans in text and assigns each a semantic type

Example sentence: “Barack Obama visited Paris in 2015 for a UNESCO event”

- A mention is a span of text, not an abstract entity in the world.
- NER usually answers two questions at once:
  - ***where*** in the text is the mention, and ***what type*** is it?
- Different datasets use different ontologies: PER/ORG/LOC/GPE/DATE/MONEY/etc.
- Some applications also need nested entities, discontinuous spans, or fine-grained labels.

Example output
(also note further category types)

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Citing high fuel prices, [ORG **United Airlines**] said [TIME **Friday**] it has increased fares by [MONEY **$6**] per round trip on flights to some cities also served by lower-cost carriers. [ORG **American Airlines**], a unit of [ORG **AMR Corp.**], immediately matched the move, spokesman [PER **Tim Wagner**] said. [ORG **United**], a unit of [ORG **UAL Corp.**], said the increase took effect [TIME **Thursday**] and applies to most routes where it competes against discount carriers, such as [LOC **Chicago**] to [LOC **Dallas**] and [LOC **Denver**] to [LOC **San Francisco**].

What counts as an entity?

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Newswire style: General domain datasets often use a small label set

PERSON | ORG | LOC | GPE | DATE | TIME | MONEY | PRODUCT

Other domains: Domain-specific systems often need richer and customised labels

- **Biomedical**: genes, proteins, diseases, chemicals
- **Legal**: statutes, courts, parties, case citations
- **Social media**: creative spellings, handles, emerging events, hashtags
- **Low-resource / local-language settings**: entity categories may need redesign rather than direct import discussion prompt from English newswire

Why does NER matter?

- NER is building block for many NLP pipelines.
- It turns raw text into structured signals for downstream use.

Information extraction

Pull key facts from reports, contracts, or articles

Search and retrieval

Improve indexing and faceted search

Question answering

Detect answer candidates and relevant passages

Knowledge graphs

Populate nodes/edges before linking and relation extraction

Analytics / monitoring

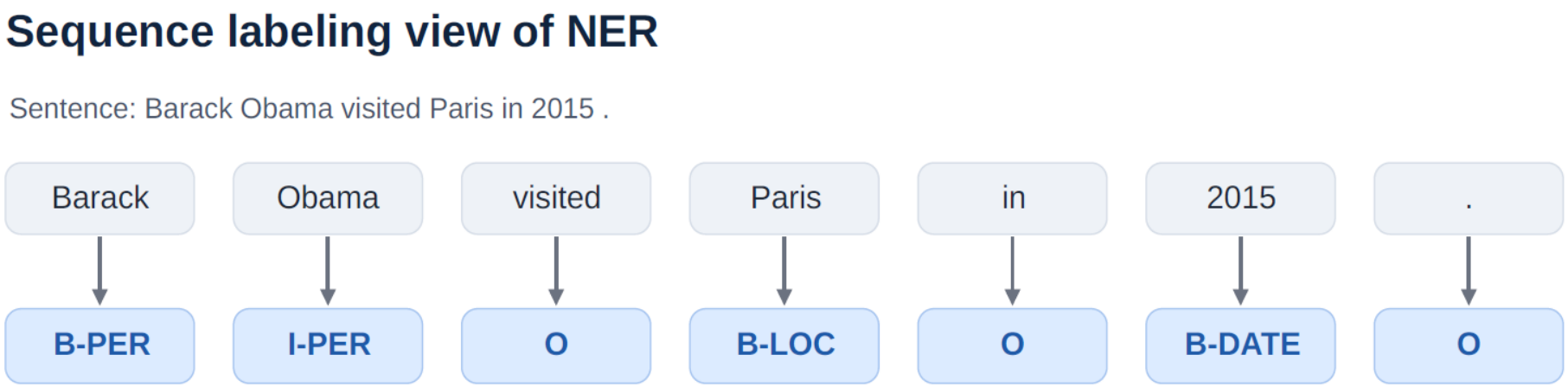
Track people, organisations, events, and locations over time

De-identification

Identify names, IDs, dates, and other sensitive fields

NER as sequence labelling

A common formulation: assign a tag to each token



- We often convert span detection into token-by-token prediction using a tagging scheme.
- The label space combines boundary information and type: B-PER, I-PER, O, etc.
- This formulation is convenient, but it assumes tokenisation decisions are already sensible.

Tagging schemes: IO, BIO and BILOU (BIOES)

Tag choices affect both training behaviour and evaluation

Jane Villanueva of United Airlines Holdings discussed the Chicago route.

Tokens	IO	BIO	BILOU	BIOES
Jane	I-PER	B-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	L-PER	E-PER
of	O	O	O	O
United	I-ORG	B-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG	I-ORG
Holdings	I-ORG	I-ORG	L-ORG	E-ORG
discussed	O	O	O	O
the	O	O	O	O
Chicago	I-LOC	I-LOC	U-LOC	S-LOC
route	O	O	O	O
.	O	O	O	O

B – *Beginning token;* **I** – *Inside token*; **O** – *Outside token* (i.e. not part of entity), **L (or E)** – *Last (End) token*; **O** – *Outside token*; **U (or S)** – Unit (or single)-token entity.

Tagging schemes: IO, BIO and BILOU (BIOES)

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A Comparison of the tagging schemes

Feature	IO	BIO	BILOU (BIOES)
Granularity	Coarse	Moderate	Fine-grained
Single-word entities	Labeled as **I-**	Labeled as **B-**	Unique tags (**U-** or **S-**)
Entity Boundaries	Ambiguous for adjacent entities of same type	Less explicit for single token	Explicitly defined start/end
Performance	Generally lowest accuracy	Standard/Good	Generally Highest (better F1 scores)
Model Convergence	Fastest (simplest)	Slower	Faster (stronger boundary signals)
Tag density (***N***= #Entity types	Lowest ($1 \times N + 1$)	Medium ($2 \times N + 1$)	Highest ($4 \times N + 1$)

Data and annotation

Most NER problems are at least partly annotation problems

- Define the **schema**:
  - what labels exist, what spans count, and how edge cases are resolved.
- Develop **annotation guidelines** with examples:
  - titles, abbreviations, coordinated names, acronyms, nested mentions.
- Measure consistency:
  - **inter-annotator agreement (IAA)** reveals where the task definition is underspecified.
- Expect disagreement on **span boundaries** and **label granularity**, not only on "hard" examples.

Boundary decisions:
Typical annotation headaches:
- Is **"University of Oxford"** one ORG span?
- Does **"the French government"** count as ORG or GPE?
- Should honorifics like **"Dr."** be inside the span?

Model – annotation mismatch:
Nested entities example: **[Bank of [England]LOC]ORG**.
Flat tagging schemes struggle here.

Key takeaway:
Data-centric message: often the biggest gains come from better guidelines, better coverage, and better error-driven relabelling.

A variety of NER methods over time …

NER has moved from hand-crafted systems to representation-heavy models

Rule-based or knowledge-based approaches

- uses gazetteers (lists of terms to recognize), dictionaries, regexes, and hand-written patterns
- fast and controllable, but brittle

Neural sequence models

- RNNs and BiLSTM-CRF models learn representations from characters and tokens,
- reduce the need for handcrafted features.

Feature-based statistical models

- Hidden Markov Models (as per POS tagging, see week 14 lecture)
- CRFs and related models rely on engineered features such as casing, prefixes, suffixes, POS tags, or neighbouring words.

Transformers and LLM-assisted methods

- dominate modern benchmarks
- use contextualised representations and transfer learning.

***In practice**, many strong systems are **hybrid**: pre-annotation with a model + human review and post-processing rules.*

Feature-based NER: the CRF era

Why conditional random fields were so influential

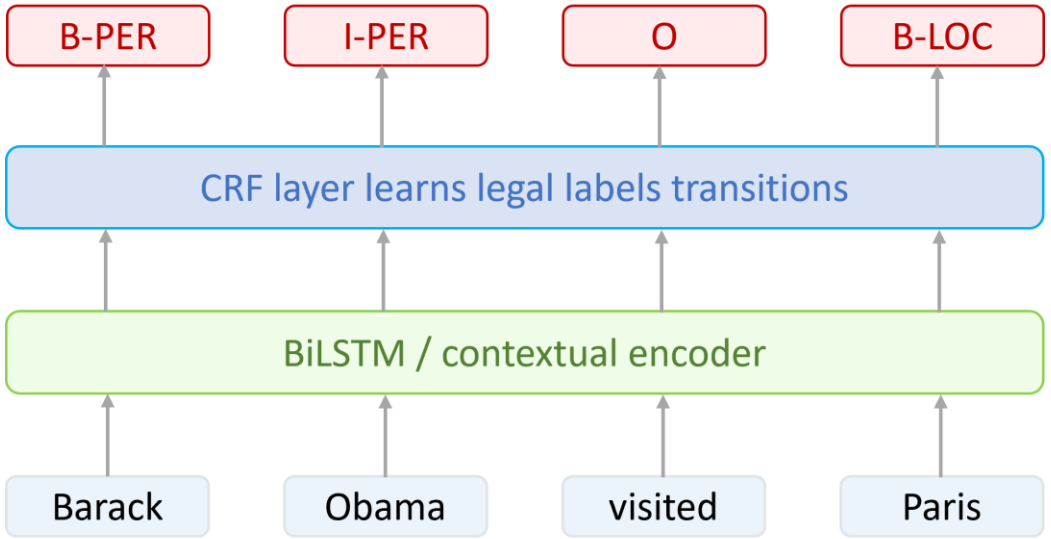
- CRFs model the ***conditional probability*** of a whole label sequence given observed features.
- They can capture ***dependencies between adjacent tags***, helping avoid illegal transitions like I-PER after B-LOC.
- **Typical features:** token shape, capitalisation, prefixes/suffixes, lexicons, POS tags, surrounding context.
- **Strengths:** interpretable features, strong with modest data, works well with carefully engineered pipelines.
- **Limitations:** manual feature work, weaker cross-domain transfer, brittle for creative spelling or morphologically rich variation.

Neural NER: BiLSTM–CRF

A strong pre-transformer architecture that still teaches useful ideas

- Character encoders help with morphology and unseen words.
- Bidirectional context captures left and right evidence.
- A CRF output layer constrains final label sequences.
- This model reduced feature engineering but still needed task-specific training data.

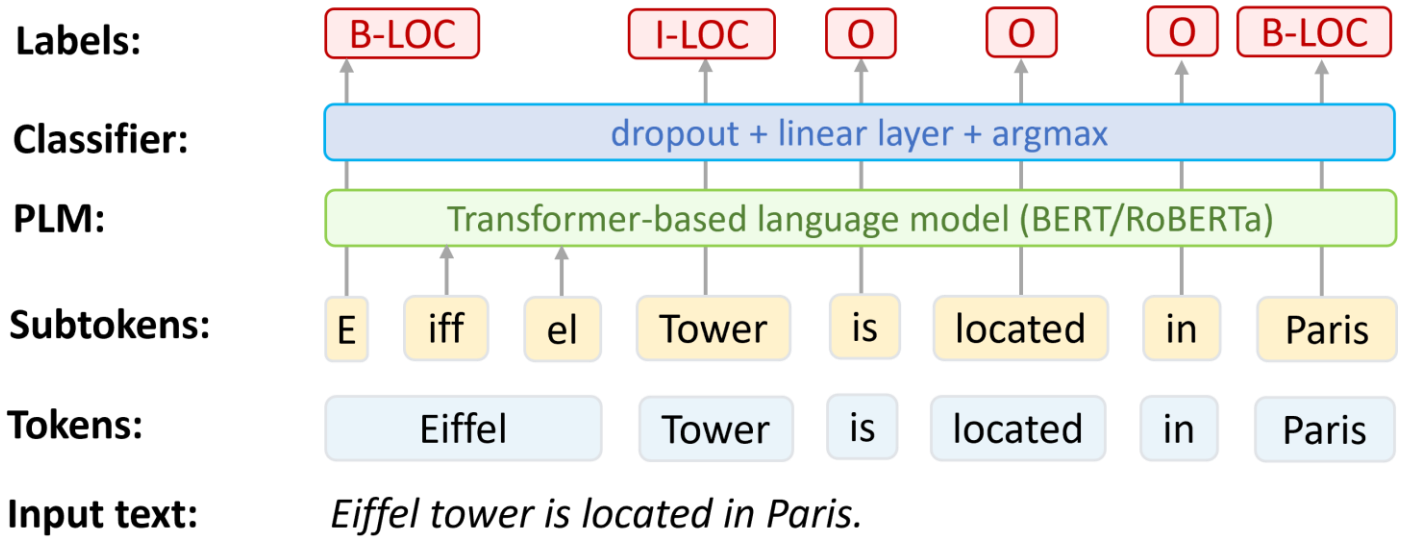
BIO Tagging Intuition:
Character/word signals → contextual encoder → structured decoder



Conceptual lesson: Representation learning + structured decoding is often stronger than either one alone.

Transformers for NER

Contextual embeddings + fine-tuning made NER much easier to transfer



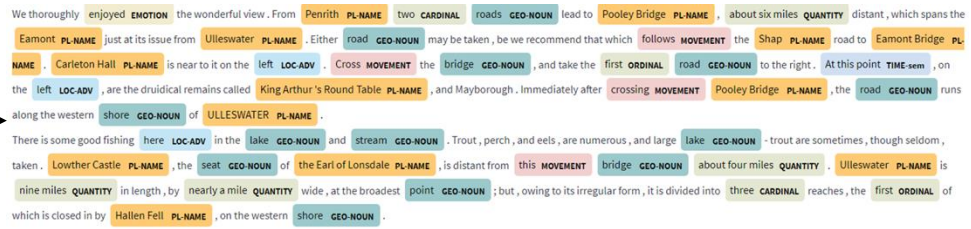
- *Typical setup:* add a token-classification head on top of a pretrained encoder (e.g., BERT-style model).
- *Main implementation detail:* word labels must be aligned to subword tokens after tokenisation. Often label the first piece and mask the rest
- Pretraining improves generalisation, but domain mismatch still hurts. Clinical, legal, or social-media text often need adapted models.

LLM-assisted and hybrid NER

Useful, but not a free replacement for supervised sequence labelling

- LLMs can perform zero-shot or few-shot extraction via prompting, often returning JSON or span lists.
- They are helpful for bootstrapping annotations, weak labelling, data cleaning, or handling long-tail labels.
- But outputs may be inconsistent, expensive, difficult to calibrate, and less reproducible than conventional taggers.
- Boundary precision can be weaker, and evaluation becomes trickier when outputs are not token aligned.
- Hybrid practice: use an LLM to suggest annotations, then train a smaller deterministic model for production.

We thoroughly enjoyed the wonderful view. From Penrith two roads lead to Pooley Bridge, about six miles distant, which spans the Eamont just at its issue from Ulleswater. Either road may be taken, be we recommend that which follows the Shap road to Eamont Bridge. Carleton Hall is near to it on the left. Cross the bridge, and take the first road to the right. At this point, on the left, are the druidical remains called King Arthur's Round Table, and Mayborough. Immediately after crossing Pooley Bridge, the road runs along the western shore of ULLESWATER.



Why NER is hard?

Common failure modes beyond "the model is bad"

Ambiguity is a problem.

- "Washington", "Apple", "Jordan", or "May" can have multiple valid types.

[$_{PER}$Washington] was born into slavery on the farm of James Burroughs.

[$_{ORG}$ Washington] went up 2 games to 1 in the four-game series.

Blair arrived in [$_{LOC}$ Washington] for what may well be his last state visit.

In June, [$_{GPE}$ Washington] passed a primary seatbelt law.

How do we label Washington [LOC, PER, ORG or GPE] in the above examples?

Why NER is hard?

Common failure modes beyond "the model is bad"

Other failure modes:

- *Variation*:
  - spelling mistakes, transliteration, abbreviations, casing noise, and code-switching.
- *Boundary difficulty*:
  - deciding whether titles, modifiers, or coordinated phrases belong inside the span.
- *Novel entities:*
  - new people, products, events, slang, or local names absent from training data.
- *Domain shift:*
  - models trained on news often degrade on clinical notes, support tickets, or social media.
- *Nested and discontinuous mentions:*
  - many real texts are not naturally flat.

Mini Exercise: What label would you assign to "*The University of Cape Town law clinic*"?

Evaluation protocols

Entity-level scoring is the usual standard

Entity-level evaluation

Gold vs system spans (exact match)

Gold spans

```
[Barack Obama]PERSON visited [Paris]LOC
```

System spans

```
[Obama]PERSON visited [Paris]LOC
```

Paris = TP (correct span + type).
Barack Obama = FN (missed). Obama = FP (wrong boundary).

Core formulas

Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2PR / (P + R)

Report exact-span + exact-type matches (CoNLL-style).
Keep token-level scores separate (they can be misleading) as most tokens have **O** tags.

Always report what counts as a match: exact span + exact type? partial span? micro-averaged across types?

Error analysis

Useful categories for analysing NER failures

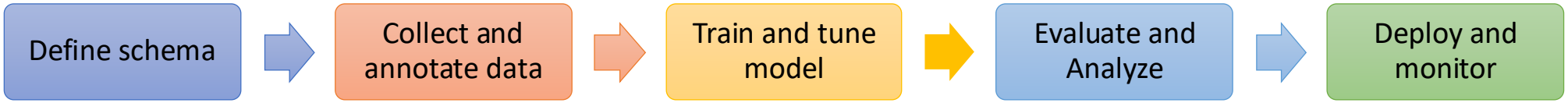
Span boundary errors | Type confusions | Missed Entities | Hallucinated spans | Domain-specific failures

- Look beyond one score: which entity types fail, and under what text conditions?
- Create an error spreadsheet with columns for gold span, predicted span, predicted type, context, and likely cause.
- Compare models qualitatively. A transformer and a CRF can have similar F1 but very different failure profiles.
- Use findings to improve the data, guidelines, postprocessing rules, or model choice.

From model to system

Production NER is a workflow, not a notebook cell

Production workflow



- Monitor drift:
  - entity types, names, and usage contexts change over time.
- Preserve provenance:
  - version the schema, guidelines, data splits, and evaluation scripts.
- Combine model outputs with human review when false negatives or false positives are costly.
- Think about latency, cost, reproducibility, and privacy when choosing model size and deployment style.
- NER is a full pipeline problem, not only a model-choice problem.

Datasets

- Semeval datasets
  - SemEval-2022 Task 11: Multilingual Complex Named Entity Recognition (MultiCoNER) https://aclanthology.org/2022.semeval-1.196/
  - SemEval-2023 Task 2: Fine-grained Multilingual Named Entity Recognition (MultiCoNER 2) https://aclanthology.org/2023.semeval-1.310/
- Manually labelled corpora
  - biomedical named entities (Bada et al., 2012),
  - literary text (Bamman et al., 2019) and Lake District travel writing (Rayson et al., 2017)

Bias, fairness, and privacy

Responsible NER requires more than a headline F1 score

- Representation bias:
  - training data may under-cover names, places, or orthographies from marginalised communities.
- Schema bias:
  - imported label sets may erase locally important categories or flatten culturally specific distinctions.
- Error asymmetry:
  - a model can score well overall while systematically failing on certain name types or dialects.
- Privacy:
  - NER is often used for de-identification, but misses can leak sensitive information.
- Human review matters when stakes are high (e.g., healthcare, legal, social services, humanitarian settings).

Mitigation:
Responsible practice ideas:
- audit by subgroup or name type
- include local annotators/domain experts
- document guidelines and known blind spots

Quiz:
For a useful de-identification tool, what performance score should be acceptable?

Any questions so far?

NER for Igbo language

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Named entity recognition for African languages : a focus on the Igbo language

Chukwuneke, CI and Rayson, Paul and Ezeani, Igna
language. PhD thesis, Lancaster University.

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Abstract

Named Entity Recognition (NER) is a crucial task f
question answering, classification, and machine tr
few other languages like Japanese, German, and F
considered under-resourced in NLP highlighting th
Named Entity Recognition (NER) for low-resource l

IGBONER 2.0: EXPANDING NAME NITION DATASETS VIA PROJECTION

Chiamaka Chukwuneke[1,2], Paul Rayson[1], Ignatius Ezeani[1]
Doris Asogwa[2], Chidimma Okpalla[3], and Chinedu Mbonu[2]

[1]Lancaster University, United Kingdom, [2]Nnamdi Azikiwe University
[3]Federal University of Technology Owerri, Nigeria

ABSTRACT

Since the inception of the state-of-the-art neural netwo
guage processing research, the major challenge faced b
is the lack or insufficiency of annotated training data. T
tion (NER) task is no exception. The need for an efficie
tation process, especially for low-resource languages ca
In this work, we leverage an existing NER tool for English in a cross-language
projection method that automatically creates a mapping dictionary of entities in
a source language and their translations in the target language using a paral-
lel English-Igbo corpus. The resultant mapping dictionary, which was manually
checked and corrected by human annotators, was used to automatically generate
and format an NER training dataset from the Igbo monolingual corpus thereby
saving a lot of annotation time for the Igbo NER task. The generated dataset was
also included in the training process and our experiments show improved perfor-
mance results from previous works.

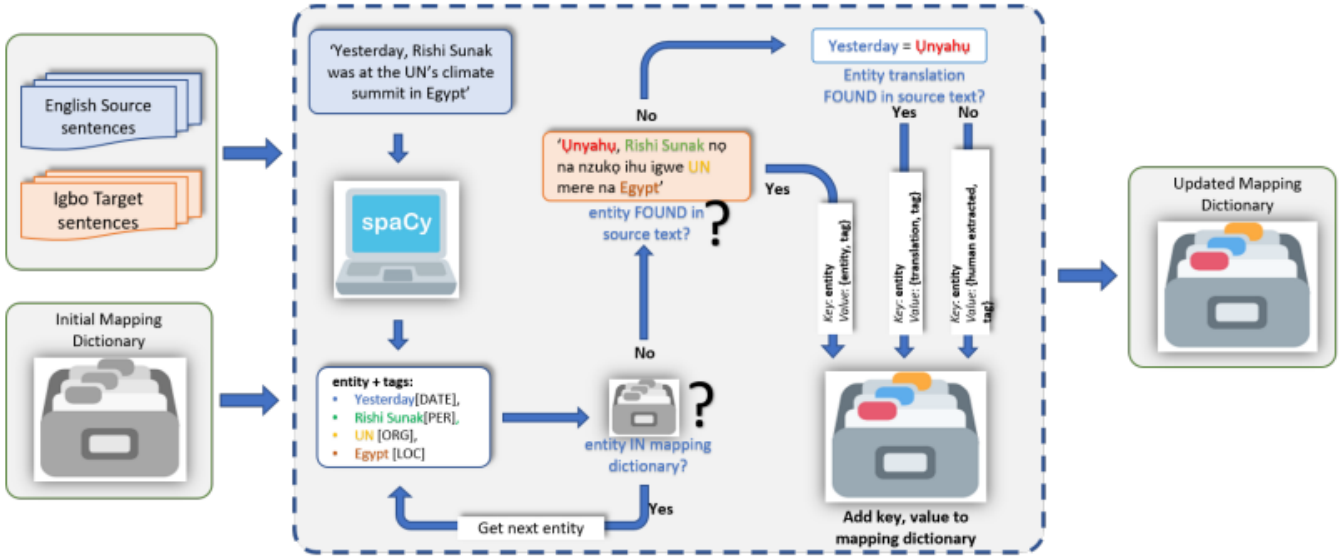

Figure 2: An illustration of the semi-automatic process used in creating the mapping dictionary.

NER for African languages

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Computer Science > Computation and Language

[Submitted on 22 Mar 2021 (v1), last revised 5 Jul 2021 (this version, v2)]

MasakhaNER: Named Entity Recognition for African Languages

David Ifeoluwa Adelani, Jade Abbott, Graham Neubig, Daniel D'souza, Julia Kreutzer, Constantine Lignos, Chester Palen-Michel, Happy Buzaaba, Shruti Rijhwani, Sebastian Ruder, Stephen Mayhew, Israel Abebe Azime, Shamsuddeen Muhammad, Chris Anuoluwapo Aremu, Catherine Gitau, Der Gwadabe, Ignatius Ezeani, Rubungo Andı David, Samba Ngom, Tosin Adewumi, Pau Chiamaka Chukwuneke, Nkiruka Odu, Eri Bateesa, Temilola Oloyede, Yvonne Wam Ayodele Awokoya, Mouhamadane MBOU Wolde, Abdoulaye Faye, Blessing Sibanda Thierno Ibrahima DIOP, Abdoulaye Diallo

We take a step towards addressing the under large publicly available high-quality dataset fo a variety of stakeholders. We detail characteri these languages pose for NER. We analyze o methods across both supervised and transfer future research on African NLP.

Computer Science > Computation and Language

[Submitted on 22 Oct 2022 (v1), last revised 15 Nov 2022 (this version, v2)]

MasakhaNER 2.0: Africa-centric Transfer Learning for Named Entity Recognition

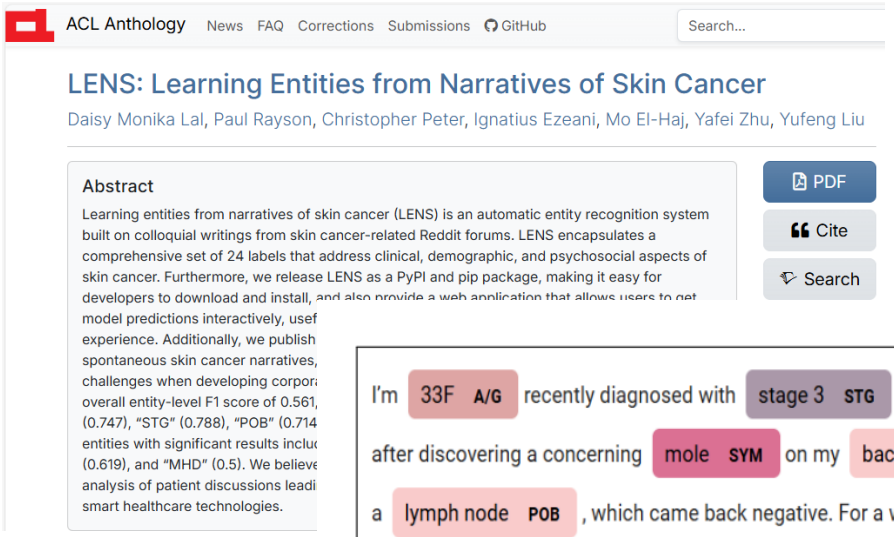
David Ifeoluwa Adelani, Graham Neubig, Sebastian Ruder, Shruti Rijhwani, Michael Beukman, Chester Palen- uhammad, Peter Nabende, Cheikh M. F. P. Dossou, Blessing Sibanda, Happy lia Taylor, Fatoumata Kabore, Chris itau, Edwin Munkoh-Buabeng, Victoire M. te, Elvis Mboning, Tajuddeen Gwadabe, eo L. Mokono, Ignatius Ezeani, Chiamaka mumin, Odunayo Ogundepo, Oreen Yousuf,

nted in NLP research and development. The tasets, as well as a lack of understanding of gress towards solutions for these challenges, st human-annotated NER dataset for 20 al transfer methods in an Africa-centric setting, mance. We show that choosing the best across 20 languages compared to using that cover typologically-diverse African

Language	Sentence
English	The Emir of Kano turbaned Zhang who has spent 18 years in Nigeria
Amharic	የካኖ ኢምር በናይጀርያ ፲፰ ዓመት ያሳለፈውን ዝንግን ዋና መሪ አደረጉት
Hausa	Sarkin Kano yayi wa Zhang wanda yayi shekara 18 a Najeriya sarauta
Igbo	Onye Emir nke Kano kpubere Zhang okpu onye nke nọgoro afọ iri na asatọ na Naijiria
Kinyarwanda	Emir w'i Kano yimitse Zhang wari umaze imyaka 18 muri Nijeriya
Luganda	Emir w'e Kano yatikkidde Zhang amaze emyaka 18 mu Nigeria
Luo	Emir mar Kano ne orwakone turban Zhang ma osedak Nigeria kwuom higni 18
Nigerian-Pidgin	Emir of Kano turban Zhang wey don spend 18 years for Nigeria
Swahili	Emir wa Kano alimvisha kilemba Zhang ambaye alikaa miaka 18 nchini Nigeria
Wolof	Emiiru Kanó dafa kaala kii di Zhang mii def Nigeria fukki at ak juróom ñett
Yorùbá	Ẹ́míà ìlú Kánò wé láwàní lé orí Zhang ẹni tí ó ti lo ọdún méjìdínlógún ní orílè-èdè Nàijíríà

Table 2: Example of named entities in different languages. PER, LOC, and DATE are in colours purple, orange, and green respectively.

NER in Medical Research



Guideline	Correct ✓	Incorrect ✗
Consistency	"*chemotherapy*" → **TRT** (consistent across the document).	"*chemotherapy*" → **TRT** in one place, "*chemotherapy*" → **MED** elsewhere.
Precision	"*stage 2 melanoma*" → "*stage 2*" → **STG**, "*melanoma*" → **CANC_T**.	"*Stage 4 melanoma*" → **CANC_T** or **STG** without separating them.
No Overlap	"*swollen lymph node*" → **SYM** (without further annotation).	"*swollen lymph node*" → **SYM** for "*swollen lymph node*" and annotating "*lymph node*" separately as **POB** (overlapping).
Punctuation Exclusion	"*33 years old,*" → annotate "33 years old" as **AGE** (excluding the comma).	"*33 years old.*" → annotating "*33 years old,*" including the period as part of the entity.
Comprehensive Coverage	"*PET scan showed mets in the brain and lungs*" → "*PET scan*" → **INV**, "*mets*" → **INV**, "brain" → **POB**, "*lungs*" → **POB**.	Missing annotations for "*brain*" and "*lungs*" as **POB**.
Reddit Slang	"*I am 33M*" or "*I am 26/F*" → "*33M*" → **A/G**, and "*26/F*" → **A/G**	Annotating "*33M*" or "*26/F*" as **AGE** → **GENDER**.
Abbreviations	"*WLE*" → **TRT** or "*BRAF/MEK*" → **MED**	Not tagging the abbreviation.

Table 3: General guidelines for LENS annotation. For abbreviations see Table 2

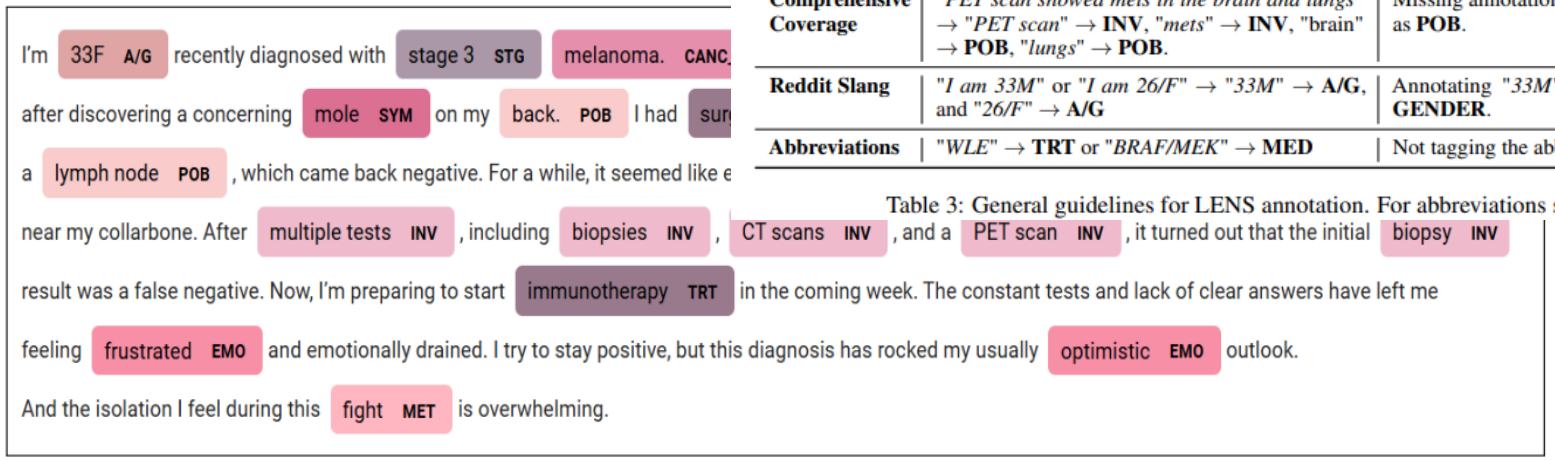
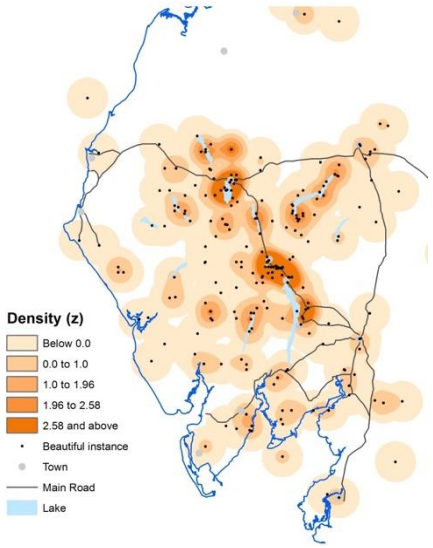


Figure 1: Extracting LENS entities from a narrative of skin cancer. For abbreviations see Table 2

NER in Spatial Narratives

- Geographical Text Analysis (GTA): combining semantic analysis (emotions and time) with Named Entity Recognition (NER)
- What themes are associated with places? What places are associated with a theme?
- Lake District corpus & Holocaust survivor testimonies

https://spacetimenarratives.github.io/
https://github.com/SpaceTimeNarratives/demo



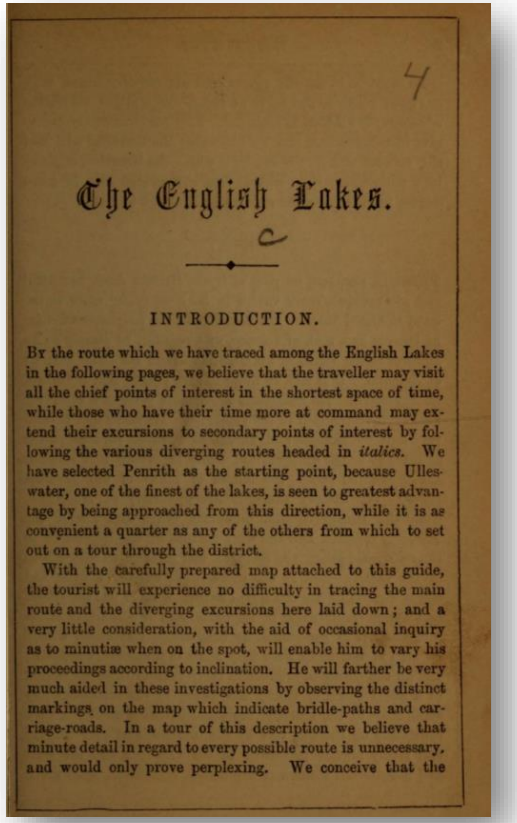
'Beautiful' places in 1.5 million words of Lake District writing

Named Entity Recognition applied to Lake District writing

From Penrith PLNAME two CARDINAL roads GEONOUN lead to Pooley Bridge PLNAME , about six miles QUANTITY distant, which spans the Eamont PLNAME just at its issue from Ulleswater PLNAME . Either road GEONOUN may be taken, be we recommend that which follows MOVEMENT the Shap road PLNAME to Eamont PLNAME Bridge. Carleton Hall PLNAME is near to it on the left MOVEMENT . Cross MOVEMENT the bridge GEONOUN , and take the first ORDINAL road GEONOUN to the right. At this point TIME-SEM , on the left MOVEMENT , are the druidical remains called King Arthur's Round Table PLNAME , and Mayborough PLNAME . Immediately after crossing MOVEMENT Pooley Bridge PLNAME , the road GEONOUN runs along the western shore GEONOUN of ULLESWATER PLNAME . To Patterdale PLNAME , a distance of ten miles QUANTITY ; but, before proceeding along it, the tourist MOVEMENT would do well to take a walk MOVEMENT of a few miles QUANTITY along the eastern shore GEONOUN , in the direction of Martindale PLNAME , from several points GEONOUN on which he will TIME-SEM obtain a good view of the lake GEONOUN .

Corpus of Lake District Writing

- Comprises 80 texts and around 1.5 million
- Texts created between 1622 – 1900
- Containing travel literature, fiction, histories, letters, and diaries.
- Includes works by well-known Lake Poets
  - e.g. William Wordsworth and Samuel Taylor Coleridge.
- Also accounts by prominent writers such as Daniel Defoe and Celia Fiennes



4

The English Lakes.

INTRODUCTION.

By the route which we have traced among the English Lakes in the following pages, we believe that the traveller may visit all the chief points of interest in the shortest space of time, while those who have their time more at command may extend their excursions to secondary points of interest by following the various diverging routes headed in *italics*. We have selected Penrith as the starting point, because Ullswater, one of the finest of the lakes, is seen to greatest advantage by being approached from this direction, while it is as convenient a quarter as any of the others from which to set out on a tour through the district.

With the carefully prepared map attached to this guide, the tourist will experience no difficulty in tracing the main route and the diverging excursions here laid down; and a very little consideration, with the aid of occasional inquiry as to minutiæ when on the spot, will enable him to vary his proceedings according to inclination. He will farther be very much aided in these investigations by observing the distinct markings on the map which indicate bridle-paths and carriage-roads. In a tour of this description we believe that minute detail in regard to every possible route is unnecessary, and would only prove perplexing. We conceive that the

The English Lakes
– by Thomas Nelson & Sons (1857)

CLDW: Lake District Corpus Repository (https://github.com/UCREL/LakeDistrictCorpus)

Holocaust Survivor’s Testimonies

- ≈1k (now 998) transcripts of survivor interviews, tokens
- Each comprises a series of interviewer's questions and the survivor’s answers
- Chronicles the survivor’s Holocaust experience
- Duration ≈ 2hrs: 20% pre-war life; 60% wartime experience; 20% post-war
- Highlights sites of incarceration or hiding serving as anchors that describe survivors’ wartime trajectories.

Key Task

Can we approach as an NER task?

With a transcript, can we extract, visualize and analyse narratives around

- Locations:
  - Names of places – cities, countries, camps – mentioned in the testimony. May be converted to coordinates for mapping e.g. Poland, Utrecht, Auschwitz-Birkenau
- Locale:
  - events that occur at a place will contribute to this
  - various forms of date, time, and other temporal references
  - as well as spatial and geographical features, ‘camp’, ‘ghetto’, ‘gas chamber’, ‘school’, ‘train’ etc. often vague or ambiguous and cannot be directly mapped in Euclidean space.
- Sense of place:
  - subjective sentiments and emotional expressions by the survivor about a place

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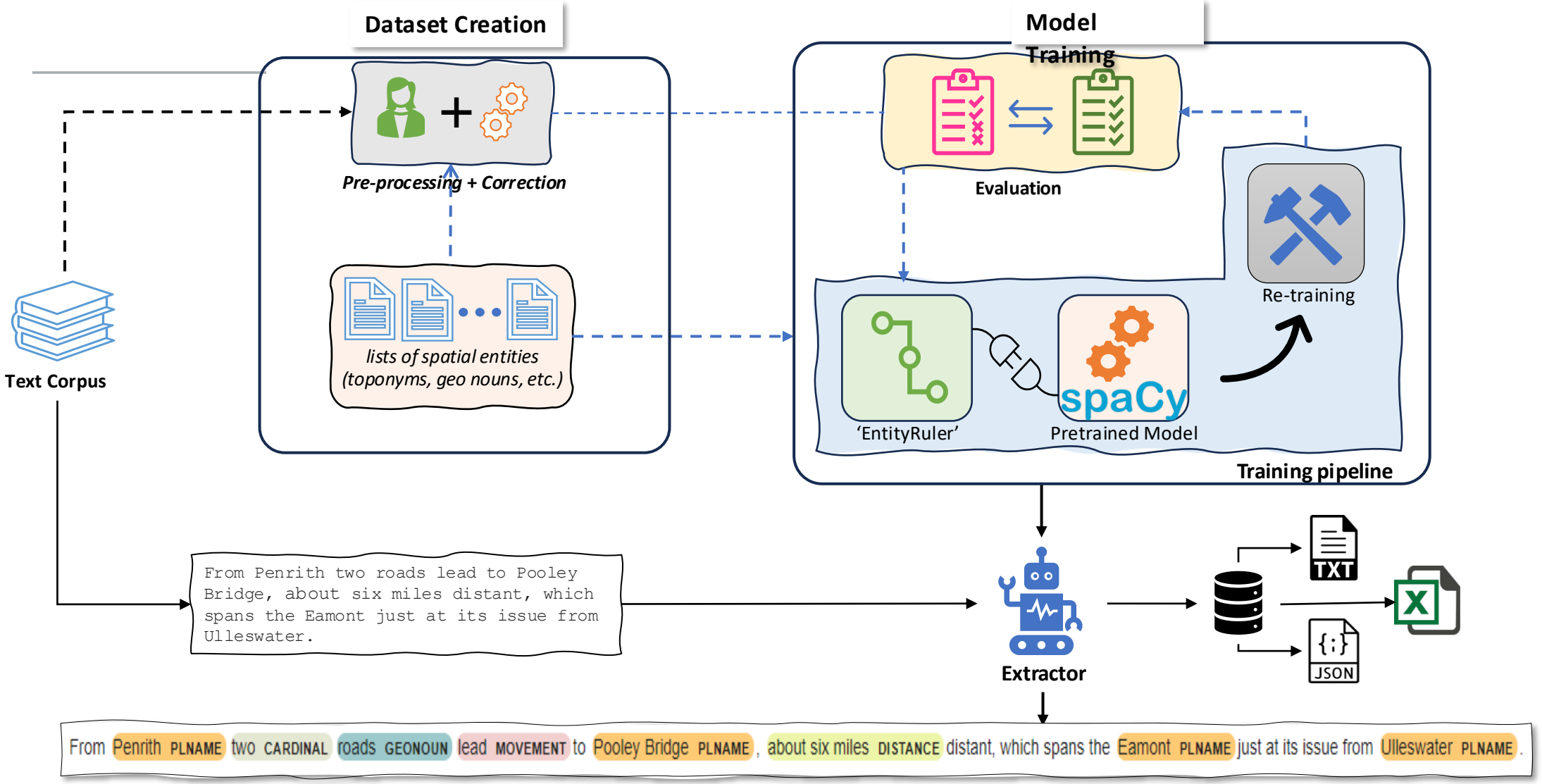
Labelling Spatio-Temporal Entities

- Names of places:
  - e.g. *Poland*:**COUNTRY**; *Utrecht*:**CITY**, *Auschwitz-Birkenau*:**CAMP**, *Europe*:**CONTINENT**
- Spatial or geographical features:
  - **GEONOUN** – concentration camp, ghetto, factory, coal mine.
- Temporal references:
  - *October 7th, 1925*: **DATE**; *the summertime*: **DATE**
  - *about 6 o'clock in the morning*-**TIME**; *less than half an hour*-**TIME**
- Events:
  - **EVENT** – *Hanukkah, World War II, Holocaust, Passover*
- Sentiments:
  - Uses a sentiment lexicon e.g. *delightful*:**+EMOTION**; *pleasant*:**+EMOTION**; *horrible*:-**EMOTION**

Annotation and Model Training



Visualizing Spatial Entities

Lancaster University

Allows different input methods

Text area

app

Old Version 0.00

Spatial Narratives Project

How do you want to input your text?

- Paste copied text
- Use an example file
- Upload data file

Select tags to visualize:

- [ ] Select|Deselect all tags
- [x] Place names: PL-NAME
- [x] Geo feature nouns: GEO-NOUN
- [x] Time: TIME|TIME-sem|DATE
- [x] Emotion: EMOTION
- [x] Movement: MOVEMENT
- [ ] Spatial Prepositions: SP-PREP
- [x] Locative Adverbs: LOC-ADV
- [x] Quantity: QUANTITY

Extracting place names and relations

Paste text to tag

We thoroughly enjoyed the wonderful view. From Penrith two roads lead to Pooley Bridge, about six miles distant, which spans the Eamont just at its issue from Ulleswater. Either road may be taken, be we recommend that which follows the Shap road to Eamont Bridge. Carleton Hall is near to it on the left. Cross the bridge, and take the first road to the right. At this point, on the left, are the druidical remains called King Arthur's Round Table, and Mayborough. Immediately after crossing Pooley Bridge, the road runs along the western shore of ULLESWATER.

There is some good fishing here in the lake and stream. Trout, perch, and eels, are numerous, and large lake-trout are sometimes, though seldom, taken. Lowther Castle, the seat of the Earl of Lonsdale, is distant from this bridge about four miles. Ulleswater is nine miles in length, by nearly a mile wide, at the broadest point; but, owing to its irregular form, it is divided into three reaches, the first of which is closed in by Hallen Fell, on the western shore.

We thoroughly enjoyed EMOTION the wonderful view . From Penrith PL-NAME two CARDINAL roads GEO-NOUN lead to Pooley Bridge PL-NAME , about six miles QUANTITY distant , which spans the Eamont PL-NAME just at its issue from Ulleswater PL-NAME . Either road GEO-NOUN may be taken , be we recommend that which follows MOVEMENT the Shap PL-NAME road to Eamont Bridge PL-NAME . Carleton Hall PL-NAME is near to it on the left LOC-ADV . Cross MOVEMENT the bridge GEO-NOUN , and take the first ORDINAL road GEO-NOUN to the right . At this point TIME-sem , on the left LOC-ADV , are the druidical remains called King Arthur 's Round Table PL-NAME , and Mayborough . Immediately after crossing MOVEMENT Pooley Bridge PL-NAME , the road GEO-NOUN runs along the western shore GEO-NOUN of ULLESWATER PL-NAME .

There is some good fishing here LOC-ADV in the lake GEO-NOUN and stream GEO-NOUN . Trout , perch , and eels , are numerous , and large lake GEO-NOUN - trout are sometimes , though seldom , taken . Lowther Castle PL-NAME , the seat GEO-NOUN of the Earl of Lonsdale PL-NAME , is distant from this MOVEMENT bridge GEO-NOUN about four miles QUANTITY . Ulleswater PL-NAME is nine miles QUANTITY in length , by nearly a mile QUANTITY wide , at the broadest point GEO-NOUN ; but , owing to its irregular form , it is divided into three CARDINAL reaches , the first ORDINAL of which is closed in by Hallen Fell PL-NAME , on the western shore GEO-NOUN .

Visualizes multiple spatial tags

https://spacetimenarratives.streamlit.app/

NER - Recap

- Named Entity Recognition (NER) is a sequence labelling task to assign entity type to words or phrases
- It is an essential task in NLP that supports other downstream tasks:
  - information extraction, question answering, search and retrieval etc
- Approaches to NER include:
  - Rule-based, feature-based, neural and transformer + LLM- based
- Evaluation, and error-analysis and constant monitoring are very essential for good performance

Further reading

- Speech and Language Processing (3rd edition 2025) chapter 17: https://web.stanford.edu/~jurafsky/slp3/17.pdf
- David Bamman, Sejal Popat, and Sheng Shen. 2019. An annotated dataset of literary entities. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), pages 2138–2144, Minneapolis, Minnesota. Association for Computational Linguistics. https://aclanthology.org/N19-1220/
- Bada, M., M. Eckert, D. Evans, K. Garcia, K. Shipley, D. Sitnikov, W. A. Baumgartner, K. B. Cohen, K. Verspoor, J. A. Blake, and L. E. Hunter. 2012. Concept annotation in the craft corpus. BMC bioinformatics, 13(1):161
- Rayson, P., Reinhold, A., Butler, J., Donaldson, C. E., Gregory, I. N., & Taylor, J. E. (2017). A deeply annotated testbed for geographical text analysis: The Corpus of Lake District Writing. In GeoHumanities'17: 1st ACM SIGSPATIAL Workshop on Geospatial Humanities: 25th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems. Association for Computing Machinery (ACM). https://doi.org/10.1145/3149858.3149865
- Zara Nasar, Syed Waqar Jaffry, and Muhammad Kamran Malik. (2021). Named Entity Recognition and Relation Extraction: State-of-the-Art. ACM Computing Surveys (CSUR). 54, 1. https://doi.org/10.1145/3445965
- Alshammari & Alanazi, 2021, The impact of using different annotation schemes on named entity recognition, Egyptian Informatics Journal, Elsevier, Science Direct

Any final questions?

Week 19 Lab Tasks (all non-assessed)

1. Further investigations into applications of semantic tagging and named entity recognition for spatial humanities (approx. 60 minutes)
   - Aim: to understand how some of the semantic field annotation and named entity recognition can be combined and applied to other research questions e.g. in the digital humanities
2. Continue with your coursework project and obtain feedback on data, experimental set up and/or drafts of the report

Task 1 Spatial Humanities applications

- In the Spatial Humanities project (https://spacetimenarratives.github.io/), we are using a combination of named entity recognition and the semantic tagger (presented in week 18) to pick out key entities and concepts to do with emotion (E), Movement, travel and transport (M), and time (T) to see how they are linked with the spatial entities in the text
- There are multiple Python notebooks and a pip installable library. Work through these demo notebooks and/or install the library and test it out on your BBC News corpus collected in week 18:
  - https://github.com/SpaceTimeNarratives/demo
  - https://github.com/SpaceTimeNarratives/spatio-textual